Higher National Diploma in Engineering (Electrical)

Fourth Year, 1st Semester Examination – 2019

EE 3204–Electrical Energy Management

Instructions for Candidates:
Answer any **Five** (05) questions.
Each question carries equal marks.

No. of questions: 06
No. of pages : 04
Time: (03) Three hours

---

1. (a). Define the following terms used in power transformer operations.
   (i).Voltage Regulations,
   (ii).Commercial efficiency/Energy efficiency. (3 marks).

   (b). (i).Briefly describe the sources of losses in transformers on load conditions. (2 marks).
   (ii).A 100kVA transformer has its maximum efficiency of 0.98 at full load unity power factor. During the day it is loaded as follows.
   8.00am to 1.00pm - 65kW , 35kVar,
   1.00pm to 6.00pm - 80kW, 50kVar,
   6.00pm to 1.00am - 30kW, 30kVar,
   1.00am to 8.00am. –no load.
   Determine the All day efficiency of transformer. (8 marks).

   (c).
   (i).Find the approximate voltage drop of a transformer when an inductive load is applied. It's primary and secondary resistances, inductive reactance are $R_1\Omega$, $R_2\Omega$,$X_1\ \Omega$, $X_2\ \Omega$ respectively. Primary and secondary currents are I_1A and I_2 A respectively and angle between secondary terminal voltage and current is φ_2
   [Hints: draw the transformer equivalent circuit as referred to secondary and relevant phasor diagram] (4 marks).

   (ii).Draw phasor diagram and connection diagram of a three phase transformer having Yd11 vector group symbol. (3marks).

2. (a). (i). What are the differences between direct lighting scheme and indirect lighting scheme. (3marks).

(ii). List down two light control methods and discuss one of them in brief. (4marks).

(iii). Give two types of sodium vapour lamps and differentiate them by considering their advantages, disadvantages and their applications. (4marks).

(b). (i). What is the method applied to vapourized the mercury in high pressure mercury vapour lamp. (2marks).

(ii). State the advantages of adding small amount of halogen vapour to the inert gas in halogen lamp. (3marks).

(iii). Briefly explain the way of emissionof light in fluorescent lamp. (4marks).

3.

(a).

(i). List down 04 desirable characteristics of a tariff. (4marks).

(ii). The flat demand rate tariff does not suitable for domestic consumers but it can use for the street lighting. Justify the answer. (3marks).

(iii). Answer **either part (1) or part (2)** stated below. Both carry equal marks.

Part (1)

A consumer has a maximum demand of 200kW at 40% load factor. If the tariff is Rs 1500/- per kW of maximum demand plus Rs 25/- per unit, find the overall cost per kWh in an annum. (4marks).

Or

Part (2)

An industrial consumer has single phase 200V supply. His monthly energy consumption is 1060kWh and maximum demand indicator at his premises indicates 30A which is charged at 0.8 power factor for 02 hrs daily at Rs 5.00 per kWh. The remaining units are charged at Rs 2.50 per kWh. Find his monthly bill.(for 30 days). (4marks).

(b). (i). What are the causes of low power factor. (3marks).

(ii). State two disadvantages of low power factor. (2marks).

(iii).A three phase 50Hz, 400V motor develops 74.6kW and the power factor being 0.75 lagging and efficiency 93%. A bank of capacitor is connected in delta across the supply terminals and power factor improved to 0.95 lagging. Determine the kVar rating of the capacitor connected in each phase. (4marks).

4. (a).(i).Why does the no load speed of three phase induction motor slow down when a mechanical load is applied on the motor shaft. (4 marks).

(ii).Show graphically the total torque develop in the three phase induction motor when rotor current lags 90^0 behind the rotor induce electromotive force. (2 marks).

(iii).What are the advantages and disadvantages of wound rotor over squirrel cage rotor. (4 marks).

(b).(i).Why the rotor iron loss in three phase induction motor is neglected under normal operating condition. (1 mark).

(ii).The power input to the rotor of a 400V, 50Hz, three phase 6 pole induction motor is 50kW. It is observed that the rotor emf makes 12 cycles per minute. Calculate,

i. Slip
ii. Rotor speed
iii. Rotor copper loss/phase
iv. Mechanical power developed
v. Rotor resistance per phase if rotor current is 50A. (5 marks).

(iii).Why does a large current drawn by the squirrel cage induction motor at the starting. (2 marks).

(iv).State the disadvantages of the voltage variation speed control method of three phase induction motors. (2 marks).

5.

(a)

(i).What do you understand by the term "Electrical Energy Management". (2 marks).

(ii).List down the four (04) duties of energy management team. (4 marks).

(b).

(i).What is the objective of carrying out energy audit. (1 mark).

(ii).State the three (03) strategies of energy audit. (3 marks).

(iii).Explain in brief over the efficient handling of electrical equipments. (3 marks).

(c). (i).What are the electrical load control techniques. (3 marks).
(ii).Why does not able to achieve 100% success DSM programme. (4 marks).

6.

(a).

(i).State the four (04) main factors which must be considered while designing the lighting scheme. (4 marks).

(ii).State five (05) factors to be considered to heating a room efficiently. (5 marks).

(b).Answer **either part (1) or part (2)** stated below, both carry equal marks.

Part (1).

A hall of 30m x 25m area with a ceiling height of 6m is to be provided with a general illumination of 225 Lumens/m^2 , taking a co efficient of utilization of 0.75 and depreciation factor 1.5.Determine the number of fluorescent tubes required, their spacing, mounting height and total wattage.

Takes luminous efficiency of fluorescent tube as 25 Lumens/Watt for 25W tube.

Draw the lamps arrangement sketch. (5 marks).

Or

Part (2).

The length, width and height of an office are 12m, 8m,and 4m respectively is to be provided with a general illumination of 500Lux, colour of ceiling is off white(r_c= 0.7),colour of walls and curtains are medium(r_w= 0.5). The distance from lamp to ceiling is 0.2m and height to the table top from the floor is 0.8m. Utilization factor and soiling rates are 0.5 and 1.5 respectively. Calculate the required number of lamp fittings to illuminate the office room Wattage of fluorescent lamp is 40W, Lamp efficiency is 70Lumens/watt. Assume 3 lamps are installed in a light fitting. (5 marks).

(c).

(i).What are the construction methods to be arranged while designing the three phase induction motor to achieve the maximum running torque. (2 marks).

(ii).Draw approximate equivalent circuit of three phase induction motor and represent all parameters in the circuit. (4 marks).